

Signalling Irregularities

Revision Guide | GERT8000-S7 | EMR Train Driver Training

1. What is a Signalling Irregularity?

Anything that causes a signal not to be displayed properly, or any failure or irregularity in the working of signalling equipment, is classed as a signalling irregularity.

The following count as signalling irregularities:

- Failure or irregularity in the working of a signal
- Irregular aspect sequence
- Missing block marker, or no signal shown when there should be one
- Aspect of a colour-light signal not being distinct or obvious
- Semaphore signal not showing correctly
- White light showing instead of red, yellow or green
- Signal or associated indicator difficult to see due to sunlight, streetlights or reflections
- Signal difficult to see due to trees, foliage or other obstructions
- A block marker or location marker that is dirty
- A failure or irregularity in the working of the on-board AWS equipment

Source: S7 s7.1

If you become aware of a signalling irregularity, you must tell the signaller IMMEDIATELY — even if you need to stop the train to do so.

2. Treating a Signal Not Showing or Not Showing Correctly

If a signal is not showing or not showing correctly, treat it as follows:

Signal type	Treat it as...
Stop signal	Being at DANGER
Distant signal	Being at CAUTION
Position-light, subsidiary or shunting signal	Being at NORMAL

This applies in any of the following situations:

- No signal is shown when there should be one
- The aspect of a colour-light signal is not clear or obvious
- There is no light at all
- A white light is showing instead of red, yellow or green
- A semaphore signal is not showing correctly
- One light is showing at a position-light or subsidiary signal when there should be two

Source: S7 s1.5

⚠ CRITICAL: If stopped or nearly stopped at a colour-light signal (that cannot display yellow) or a semaphore signal at danger, and it then clears — be prepared to stop at the next stop signal worked by the same signal box.

Source: S7 s1.6

3. Reporting: When to Stop and When to Wait

There are two categories of irregularity depending on urgency:

Report IMMEDIATELY — stop specially if necessary	Report at first convenient opportunity
Failure or irregularity in the working of any signal	Signals becoming difficult to see (foliage/obstructions — "becoming", not yet "difficult")
Irregular aspect sequence	Position-light, subsidiary or shunting signals that are irregular but apply to another line
White light instead of red/yellow/green	AWS fault where the failure is to give a clear indication (unless fault code 5 or 7 — see below)
Semaphore signal not showing correctly	
Signal aspect not distinct or obvious	
Signal difficult to see due to sunlight or street lights (already difficult)	
Block marker/location marker that is dirty	

Source: S7 s7.1

For signals that apply to another line — position-light, subsidiary or shunting signals — you do not need to stop specially. Tell the signaller at the first opportunity without causing delay.

4. Signals Difficult to See

4.1 Sunlight, Street Lights or Reflections

If a signal is difficult to see because of sunlight, street lights or reflections, tell the signaller immediately. You do not need to stop specially unless the signal cannot be identified.

Source: S7 s7.3

4.2 Trees, Foliage or Other Obstructions — Already Difficult

If a signal is already difficult to see because of trees, foliage or other obstructions, tell the signaller at the first convenient opportunity. You do not need to stop specially.

Source: S7 s7.4

4.3 Signals Becoming Difficult to See

If a signal is becoming (not yet fully) difficult to see because of growing foliage, tell the signaller at the first convenient opportunity. The signal does not need to be treated as defective yet.

Source: S7 s7.5

Key distinction: "already difficult to see" (sunlight) → report immediately. "Becoming difficult to see" (foliage growing) → report at first convenient opportunity, no need to treat as defective.

5. Lineside Boards, Indicators and Signs

Tell the signaller IMMEDIATELY if any of the following is missing, difficult to see, or unlit when it should be lit:

- A limit of shunt signal or indicator
- A stop board

- A shunt entry board
- Any other lineside board or sign

Source: S7 s7.2

6. Train Signalled Towards a Wrong Route

If a train has been signalled towards a wrong route, you must:

- Stop the train as soon as it is possible to do so safely
- Tell the signaller

Source: S7 s1.2

If the signal has been cleared for a wrong route:

- Stop the train at the signal if it is possible to do so safely
- Tell the signaller — they will replace the signal to danger so the correct route can be set
- If the train has not been able to stop before the signal, it may be necessary for a wrong-direction movement — but only when authorised by the signaller

If an electric train is wrongly routed onto a non-electrified section, take great care regarding the electrical equipment (pantographs and collector shoes) before proceeding.

Source: S7 s1.2 / S5 incorrect routing

7. Starting After Stopping at or Near a Signal

7.1 Stopped on the Approach to a Signal Showing Proceed

If you have to stop on the approach to a signal that is showing proceed, make sure it still shows proceed before you re-start the train. If the train cannot continue, tell the signaller immediately.

Source: S7 s2.1

7.2 Stopped Before the Whole Train Has Passed a Signal Showing Proceed

If you have stopped before the whole train has passed a signal that was showing proceed, you may act on the aspect that was being displayed when you passed the signal — unless you are instructed that the train must not proceed.

Source: S7 s2.2

8. Position-Light, Subsidiary and Shunting Signals

8.1 Passenger Trains

You must not proceed with a passenger train on the authority of a semaphore shunting signal or a position-light signal alone — unless authority is published or you are instructed by the signaller.

Exception: you may proceed on the authority of a position-light or semaphore subsidiary signal associated with a main aspect signal when entering a permissive platform line.

Source: S7 s3.1

8.2 Route Indication Not Shown

If a position-light or subsidiary signal is cleared but the normal route indication is not shown:

- Make sure the movement is made at caution
- Be prepared to stop before reaching any obstruction

Source: S7 s3.2

8.3 Returning to the Approach Side of a Signal

After a shunting movement on the authority of a position-light or shunting signal, you must not proceed on your journey until:

- The movement has returned to the approach side of the signal, AND
- The signal displays the appropriate proceed aspect for the movement

Source: S7 s3.3

9. Reverse Movements

A train or shunting movement may only be reversed when one of the following applies:

- The signal controlling the movement is cleared
- The signaller gives permission to move towards a signal which will control the further movement
- The leading end is beyond the signal and it cannot be cleared — proceed under s4.2
- There is no signal and the signaller gives permission for a wrong-direction movement

Source: S7 s4.1

9.1 Train Standing Beyond a Signal

When the signal CAN be cleared: no part of the train must move until the signal is cleared.

When the signal CANNOT be cleared:

- Find out whether a movement can return the whole train to the approach side of the signal
- If that is not possible, ask the signaller for permission to proceed beyond the signal in the direction to which it applies

Source: S7 s4.2

10. AWS Irregularities

10.1 Cancelling an AWS Warning

You must immediately cancel each warning indication and obey the signal aspect or control speed to the limit shown on the warning board. If you fail to cancel in time and the brakes apply automatically:

- Make sure the train comes to a stand
- Tell the signaller what has happened

If both you and the signaller are sure it was not TPWS on-track equipment that caused the brake application, the train can proceed normally.

Source: S7 s5.1

10.2 AWS Warning When a Semaphore Distant Shows Clear

Treat the semaphore distant signal as being at caution if you receive an AWS warning when the signal is showing clear — UNLESS:

- The signal changed to clear after the train passed over the AWS magnet, OR
- A warning board or emergency indicator is positioned at the signal

Source: S7 s5.2

10.3 AWS Warning With No Track Equipment

If you receive an AWS warning and are certain the train has not passed over any AWS on-track equipment:

- Proceed normally
- Report it to the signaller at the earliest opportunity

Source: S7 s5.3

10.4 AWS Faults to Report IMMEDIATELY (Fault Codes)

Stop the train specially if necessary to tell the signaller immediately if:

- You receive an AWS CLEAR indication when a WARNING should have been received (fault code 5)
- You receive NO AWS indication when a WARNING indication should have been received (fault code 7)

All other AWS faults where the failure is to give a clear indication must be reported at the first convenient opportunity.

Source: S7 s7.8d / RS522

Fault codes 5 and 7 are the most safety-critical — they mean you got "all clear" or nothing when you should have been warned about a caution ahead. Always stop specially if needed.

11. TPWS

11.1 Automatic Brake Application (TPWS Activation)

If TPWS initiates an automatic brake application, you must:

- Acknowledge the TPWS brake demand
- Make sure the train comes to a stand
- Tell the signaller what has happened
- Carry out the signaller's instructions
- Not make any further movement until instructed

Source: S7 s6.1

11.2 TPWS at Buffer Stops

If TPWS activates when approaching buffer stops:

- Acknowledge the TPWS brake demand
- After the train comes to a stand, move forward to the normal stopping point if safe to do so
- Tell the signaller what has happened and carry out their instructions

Source: S7 s6.2

11.3 TPWS Train Stop Override

You must only use the TPWS train stop override when authorised in the rules — specifically, when authorised to pass a single signal at danger (other than the circumstances where temporary isolation applies).

Source: S7 s6.4 / S5 s3.4

11.4 TPWS Temporary Isolation

Temporary isolation (not just override) is only permitted when entering:

- A section where extended block working is in operation
- A single-line section where pilotman or modified working is in operation AND more than one signal at danger must be passed
- A line under possession

You **MUST** reinstate TPWS before leaving that section.

Source: S7 s6.3 / S5 s3.4

12. Completing Form RT3185

RT3185 must be completed by the driver (and signaller) for any signalling irregularity or failure — in addition to informing the signaller.

You do NOT need to complete RT3185 immediately if the signaller can confirm the fault is clearly a right-side failure, or can explain why it is and confirm the circumstances. You must still complete it at the first convenient opportunity.

Source: S7 s7.8

RT3185 should be carried as part of your personal equipment at all times.

13. Right-Side vs Wrong-Side Failures

Type	Definition
Right-side failure	The system fails safe — e.g. a signal shows danger/caution when it should show clear. Safer outcome.
Wrong-side failure	The system fails dangerously — e.g. a signal shows clear when it should show danger. Critical risk.

You must know the difference to make a correct assessment of whether you need to stop. A wrong-side failure can never be excused — always treat the signal appropriately (see section 2 above).

Questions & Answers

Q1: You notice a signal is not showing any aspect. How must you treat it?

A: Treat it as a stop signal at danger. If a distant signal shows nothing, treat it as being at caution. Position-light/shunting signals with no aspect — treat as being at normal.

Source: S7 s1.5

Q2: You observe a colour-light signal showing a white light instead of red, yellow or green. What must you do?

A: Treat it as a stop signal at danger. Tell the signaller immediately — stop specially if necessary. Complete RT3185.

Source: S7 s1.5 / s7.1 / s7.8

Q3: You report that a signal is difficult to see because of sunlight. What happens next?

A: The signaller tells Operations Control, tells the driver of the next approaching train what has happened, and asks them to report back on the state of the signal. Trains are signalled normally in the meantime.

Source: S7 s7.3

Q4: A signal is becoming difficult to see because of growing foliage, but is still readable today. When must you report it?

A: At the first convenient opportunity. You do not need to stop specially. The signal does not need to be treated as defective.

Source: S7 s7.5

Q5: You receive an AWS warning indication when you can see the semaphore distant signal is already showing clear. What must you do?

A: Treat the signal as being at caution, unless the signal cleared after the train passed over the AWS

magnet, or a warning board is positioned at the signal.

Source: S7 s5.2

Q6: You receive an AWS clear indication when approaching a signal that should have given you a warning. What fault code is this and what must you do?

A: This is fault code 5. You must immediately tell the signaller — stopping the train specially if necessary. Complete RT3185.

Source: S7 s7.8d / RS522

Q7: You receive no AWS indication at all when approaching a signal that should have warned you. What is this fault code and what must you do?

A: This is fault code 7. Immediately tell the signaller — stopping specially if necessary. Complete RT3185.

Source: S7 s7.8d / RS522

Q8: TPWS activates and applies the brakes. List your immediate actions.

A: Acknowledge the TPWS brake demand. Make sure the train comes to a stand. Tell the signaller what happened. Carry out their instructions. Do not make any further movement until instructed.

Source: S7 s6.1

Q9: When is TPWS temporary isolation (not just override) permitted?

A: Only when entering a section of line where extended block working is in operation; or a single-line section with pilotman/modified working where more than one signal at danger must be passed; or a line under possession. TPWS must be reinstated before leaving the section.

Source: S7 s6.3 / S5 s3.4

Q10: You stop before the whole train has passed a signal that was showing proceed. The signal is now at danger. May you proceed?

A: Yes — you may act on the aspect that was being displayed when you passed the signal, unless you are specifically instructed that the train must not proceed.

Source: S7 s2.2

Q11: A signalling irregularity applies to a position-light signal on an adjacent line (not your line). When must you tell the signaller?

A: At the first opportunity without causing delay. You do not need to stop specially for signals on another line — but main aspect signals on any line must still be reported immediately.

Source: S7 s7.1

Q12: What is a "wrong-side failure" and why does it matter?

A: A wrong-side failure is when the signalling system fails dangerously — for example, showing a clear aspect when danger should be displayed. It matters because the driver receives no warning and may not stop. A right-side failure (showing danger when clear) is safe — just inconvenient.

Source: S7 / PPT 33-05